Strong Convergence in the Finite Coxeter Numbers Game: Parabolic Length and the Complete Boundary Atlas

Route-A source-local certificate HCS-C286
Revision round 0

2 September 2026

Abstract

We fix a finite reduced crystallographic root system and fire a simple node only when its coroot coordinate is strictly positive. In the strictly dominant chamber, every legal play is a reduced word for the longest Weyl element and has $|\Phi^+|$ moves.

1 Finite owner and firing convention

Let $\Phi \subset V$ be a finite reduced crystallographic root system, with simple roots $\{\alpha_i : i \in I\}$, coroots $\alpha_i^\vee$, positive roots Φ^+, and Weyl group $W = \langle s_i : i \in I\rangle$. We use

$$A_{ij} = \langle \alpha_j, \alpha_i^\vee \rangle. \tag{1}$$

For a dominant weight λ, write

$$x_i = \langle \lambda, \alpha_i^\vee \rangle \geq 0. \tag{2}$$

Node i is legal precisely when $x_i > 0$. Firing it replaces λ by $s_i\lambda$ and therefore

$$x'_j = x_j - A_{ji}x_i. \tag{3}$$

A firing word $(i_1, \ldots, i_m)$ accumulates on the left:

$$w_m = s_{i_m} \cdots s_{i_1}, \qquad \lambda_m = w_m\lambda. \tag{4}$$

Equations (1)–(4) remove the sign, transpose, and word-order ambiguities that otherwise obscure comparisons between versions of the numbers game.

The game and its strong-convergence theory are classical. Mozes introduced reflection processes on graphs and Weyl groups [5]; Eriksson developed the Coxeter formulation and strong-convergence theorem [2, 3]. We claim no priority. Our purpose is to give a convention-locked finite-type proof, close every dominant boundary, and make the reconstruction independently executable. Standard finite Coxeter and root-system facts are sourced to Humphreys and Bourbaki [4, 1].

2 The reflection-sign mechanism

Let ℓ denote Coxeter length. The following observation is the local engine of the proof.

Lemma 1 (root-sign criterion)**.** *Suppose the current position is $w\lambda$. Then node i is legal if and only if $w^{-1}\alpha_i$ is a positive root whose support is not contained in $J = \{j : \langle \lambda, \alpha_j^\vee \rangle = 0\}$. Every legal firing obeys*

$$\ell(s_iw) = \ell(w) + 1. \tag{5}$$

Proof. The current coordinate is

$$\langle w\lambda, \alpha_i^\vee \rangle = \langle \lambda, w^{-1}\alpha_i^\vee \rangle. \tag{6}$$

A positive coroot is a nonnegative integral combination of simple coroots. Its pairing with λ is zero exactly when the corresponding root lies in the subsystem $\Phi_J = \Phi \cap \operatorname{span}\{\alpha_j : j \in J\}$; outside that subsystem it is positive. A negative root gives a nonpositive pairing. This proves the legality criterion. The standard root criterion for Coxeter length says $w^{-1}\alpha_i > 0$ exactly when (5) holds. ☐

Theorem 1 (strict chamber)**.** *If $x_i > 0$ for every i, every complete legal firing sequence terminates at $w_0\lambda$, accumulates to the longest element w_0, and has exactly $|\Phi^+|$ moves.*

Proof. Here $J = \varnothing$. Lemma 1 makes every legal word reduced, so length strictly increases and termination follows from finiteness of W. At a terminal position every simple coordinate is nonpositive; hence the position is anti-dominant. A finite Weyl orbit meets the closed anti-dominant chamber in exactly one point, here $w_0\lambda$. Strict dominance makes the stabilizer trivial, so the accumulated element is w_0. Finally $\ell(w_0) = |\Phi^+|$. □

Conclusion

Positive-coordinate firing on a finite crystallographic Weyl system is an exact parabolic quotient ascent. In the strict chamber this reaches w_0.

References

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